DIC12
Delay Elements and Tapered Buffers

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Follow Lab 12 (PDF pages 52–54) to compare size-dependent delay in five-stage chains and evaluate taper factors for five stages driving 2 pF.

2 Specification mapping

Handout requirement (PDF page)	Implementation	Verification and scope
Step 1: all four Table 12.1 sizes (52,54)	01_delay_elements	Four parallel five-stage chains
Step 2: change taper n (53–54)	02_taper_buffer	Five stages, n=2/3/4/5, C_L = 2 pF

3 Circuit and design derivation

The first part uses all four numerical W_p, W_n, L entries in Table 12.1. Case 2 lists $W_p = 1.2\,\mu$m, while the prose calls it half of case 1 (which would be 1.8). This implementation follows the table and records the inconsistency. The altered dimensions and ratios in the reference code are restored to the table. For the taper, stage k widths equal the first-stage widths times n^{k-1}. Length is fixed at $0.25\,\mu$m and first-stage $W_n/W_p = 0.6/1.8\,\mu$m. The final 2 pF load corrects the reference code's 1 pF.

The estimate $t \sim RC$ explains the tradeoff: wider devices reduce channel resistance but add self and downstream capacitance, so total delay need not decrease monotonically.

4 Simulation setup

This teaching rewrite uses a separately supplied model at 1.0 V and 25 °C. The specified topology, loads and comparative dimensions are retained. The handout's 2.5 V process results are not acceptance values for this model. The chosen 1 V is a simulation condition, not a verified device rating. MOS bulk terminals are explicitly connected to the corresponding supply rails.

HSPICE S-2021.09 uses RUNLVL=6, retaining waveforms, printed data and native measurements. Figures are redrawn from actual data; CSV files are in each Lab's results directory. Successful solution does not establish process reliability. Warnings from encrypted model blocks remain suppressed; the existing diagnostic did not expose actionable further details, so the warnings are not declared harmless and complete model-limit compliance is not claimed.

Delay chains use 0.1 ns input edges, a 5 ns period and 15 ns simulation. The taper uses the same edge rate, a 2 ns initial delay, 20 ns high interval, 45 ns period and 90 ns duration to allow settling of the large load. Delays run from the input 50% crossing to the opposite-polarity stage-5 50% crossing; both t_{pHL} and t_{pLH} are reported.

5 Results and specification checks

All five parameter cases completed normally. Dimensions are in μm and delays in ns.

Case	W_p	W_n	L	t_{pHL}	t_{pLH}
1	3.6	1.2	0.25	0.3634	0.3786
2	1.2	1.2	0.25	0.3515	0.3926
3	3.6	1.2	0.5	1.2781	1.3400
4	1.8	1.2	0.5	1.2200	1.3434

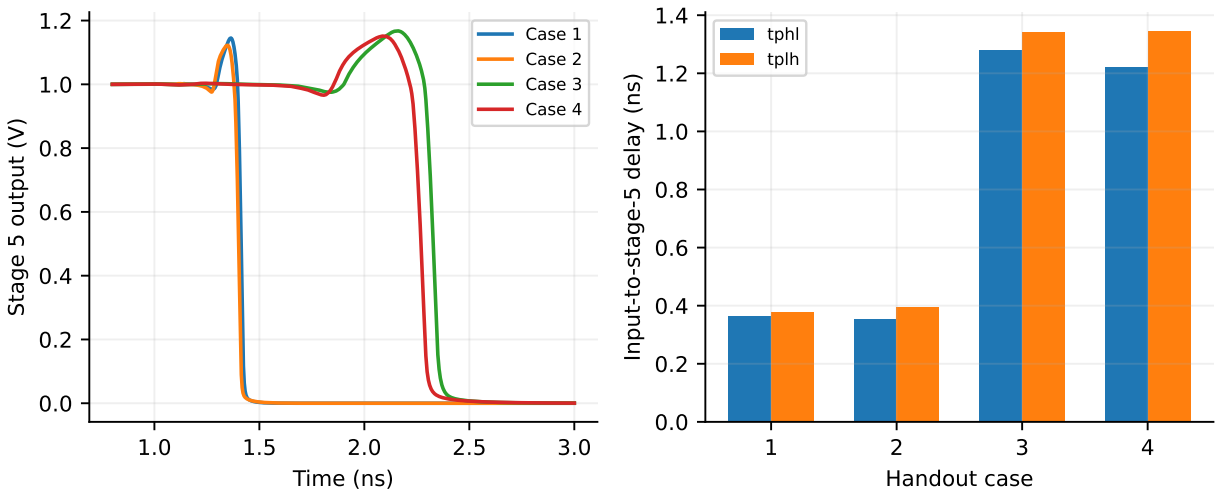


Figure 1: Results for the four delay-element sizes; redrawn from this run's HSPICE data.

n	t_{pHL} (ns)	t_{pLH} (ns)	Mean (ns)
2	0.9314	1.1852	1.0583
3	0.8779	0.9877	0.9328
4	0.9924	1.3381	1.1652
5	1.5245	2.2524	1.8884

6 Discussion

Longer channels substantially increase delay, while reduced PMOS width affects the two transition directions differently. Cases 1 and 2 have nearly equal mean delays in this model, so the handout's

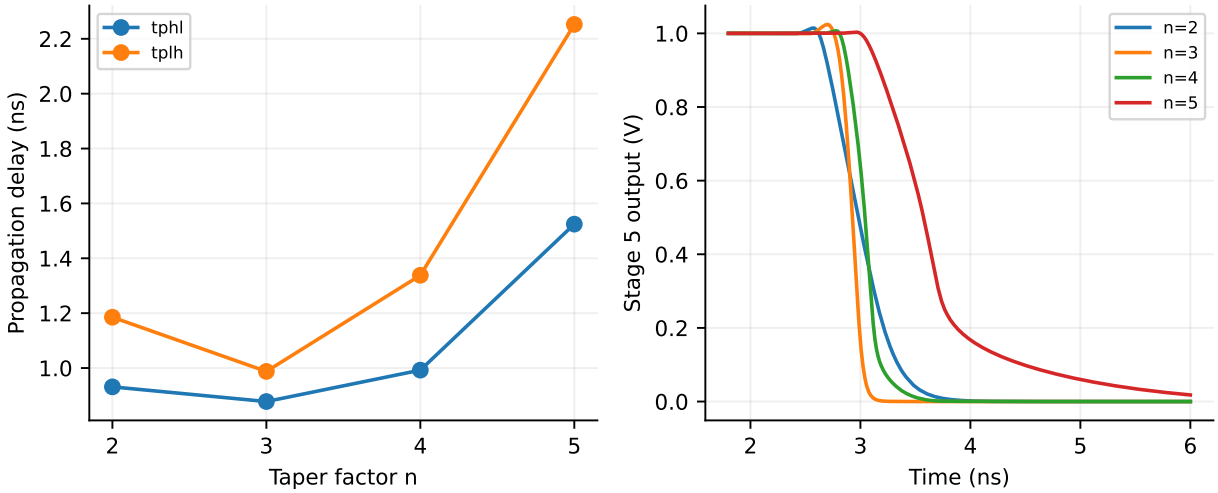


Figure 2: Taper comparison at five stages and 2 pF; redrawn from this run's HSPICE data.

single-delay ordering is not universal. Among n=2–5, n=3 gives the smallest mean delay, about 0.933 ns. This matches the handout's best-factor trend, not its 0.449 ns value. Other stage counts, non-geometric sizes and loads were not searched, so global optimality is not claimed.

7 Conclusion

Both steps are covered with explicit resolution of the table/prose conflict, the specified 2 pF load and five-stage topology. Conclusions are limited to the tested parameter set.

8 MOS transistor schematics

Source: `01_delay_elements.sp` and `02_taper_buffer.sp`. Each stage is the MOS inverter below. The first deck compares four independent five-stage chains, with (W_n, W_p, L) in μm equal to $(1.2, 3.6, 0.25)$, $(1.2, 1.2, 0.25)$, $(1.2, 3.6, 0.5)$ and $(1.2, 1.8, 0.5)$. Those chains have no added output capacitor.

The drawings use the device and node names of the cited netlists. Equal net labels denote an electrical connection. Unless stated otherwise, NMOS bulk terminals connect to ground and PMOS bulk terminals connect to V_{DD}. The bulk wires are omitted from the three-terminal symbols for readability.

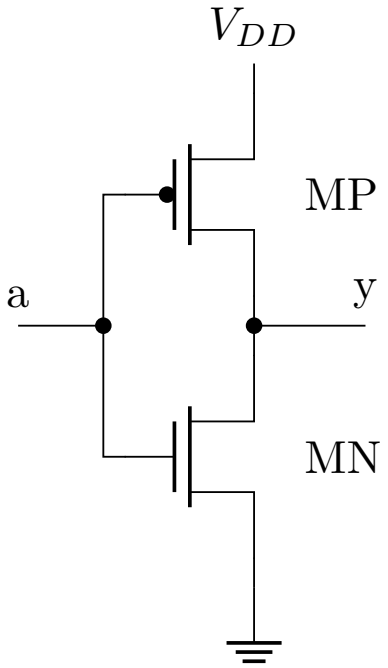


Figure 3: Two-MOS INV cell used in every delay stage. The named sizing parameters are wn, wp and ll.

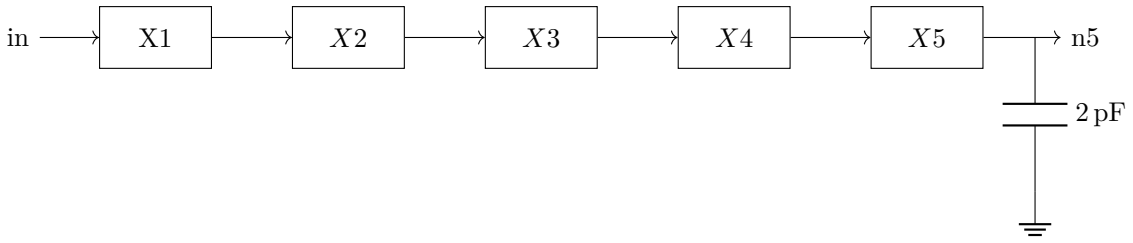


Figure 4: Tapered five-stage buffer in the second deck: stage i uses $W_n = 0.6F^{i-1}$ and $W_p = 1.8F^{i-1}\,\mu$m, $L = 0.25\,\mu$m; $F = 2, 3, 4, 5$. Only n5 drives the 2 pF load.